FOP - Lab - Task 4

Task 4.1:

Source code:

14:10:26 > task1.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
        float centiGrade
    float centiGrade;
    float fahrenHeit;

    cout << "Enter temperature in Centigrade: ";
    cin >> centiGrade;

    cout << centiGrade << " Degrees Centigrade is equal to " << (centiGrade*9/5) + 32 << " Fahrenheit.\n";

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

rehanshahid@Muhammads-MacBook-Air 14:10:26 % g++ task1.cpp -o task1
rehanshahid@Muhammads-MacBook-Air 14:10:26 % ./task1
Enter temperature in Centigrade: 37
37 Degrees Centigrade is equal to 98.6 Fahrenheit.
rehanshahid@Muhammads-MacBook-Air 14:10:26 %
```

Task 4.2:

Source code:

14:10:26 > task2.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{

    int n;

    cout << "Enter a 3 digit number: ";
    cin >> n;

    cout << "There are " << (n/100)<< " Hundreds, " << (n/10) - ((n/100)*10) << " Tens and " << n - ((n/10)*10) << " Ones in " << n << "\n";


    return 0;
}

/*
123
1   (123/100) as we declared n as integer than it must give integer value ignoring decimals
2   (123/10) - ((123/100)*10)
3   (123) - ((123/10) * 10)

*/
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

rehanshahid@Muhammads-MacBook-Air 14:10:26 % g++ task2.cpp -o task2
rehanshahid@Muhammads-MacBook-Air 14:10:26 % ./task2
Enter a 3 digit number: 547
There are 5 Hundreds, 4 Tens and 7 Ones in 547
rehanshahid@Muhammads-MacBook-Air 14:10:26 %
```

Task 4.3
Source code:

14:10:26 > task3.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{

    float firstN, secondN;

    cout << "Enter the first number: ";
    cin >> firstN;

    cout << "Enter the second number: ";
    cin >> secondN;

    cout << firstN << " is equal to " << (firstN/secondN)*100 << "% of " << secondN << ".\n";

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

rehanshahid@Muhammads-MacBook-Air 14:10:26 % g++ task3.cpp -o task3
rehanshahid@Muhammads-MacBook-Air 14:10:26 % ./task3
Enter the first number: 30
Enter the second number: 40
30 is equal to 75% of 40.
rehanshahid@Muhammads-MacBook-Air 14:10:26 %
```

Task 4.4
Source code:

```cpp
#include <iostream>
using namespace std;

int main()
{

    int n;

    cout << "Enter a number: ";
    cin >> n;

    //print results
    cout << n << " + 5 = "; // It is preffered to use more than one cout when you are incrementing or decrementing a variable in order because cout can be implemented in random o
    cout << ++++++++++n << "\n";

    cout << ----------n << " - 3 = ";
    cout << ------n << "\n";

    cout << "(" << ++++++n << " + 3) - 2 = ";
    cout << ++++++----n << "\n";

    cout << "((" << --n << " + 5) *2 / (";
    cout << n << " + 3)) = ";
    cout << (++++++++++n * 2) / (----------++++++n) << "\n";

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

rehanshahid@Muhammads-MacBook-Air 14:10:26 % g++ task4.cpp -o task4
task4.cpp:24:22: warning: multiple unsequenced modifications to 'n' [-Wunsequenced]
   24 |     cout << (++++++++++n * 2) / (----------++++++n) << "\n";
      |                      ^          ~~
1 warning generated.
rehanshahid@Muhammads-MacBook-Air 14:10:26 % ./task4
Enter a number: 20
20 + 5 = 25
20 - 3 = 17
(20 + 3) - 2 = 21
((20 + 5) *2 / (20 + 3)) = 2
rehanshahid@Muhammads-MacBook-Air 14:10:26 %
```

It shows sequential error due to many increments and decrements but execution file runs.

Post Lab Task

Source code:

```cpp
#include <iostream>
using namespace std;

int main()
{

    int a , b;

    cout << "Enter integer a: ";
    cin >> a;
    cout << "Enter integer b: ";
    cin >> b;

    //print results
    cout << "(" << a << " + " << b << ")^2 = " << a*a + 2*a*b + b*b << "\n";
    cout << "(" << a << " - " << b << ")^2 = " << a*a - 2*a*b + b*b << "\n";
    cout << "(" << a << " + " << b << ")^3 = " << a*a*a + 3*a*a*b + 3*a*b*b + b*b*b << "\n";
    cout << "(" << a << " - " << b << ")^3 = " << a*a*a - 3*a*a*b + 3*a*b*b - b*b*b << "\n";
    cout << a << "^2" << " - " << b << "^2 = " << (a-b) * (a+b) << "\n";
    cout << a << "^3" << " - " << b << "^3 = " << (a-b)*(a-b)*(a-b) + 3*a*b*(a-b) << "\n";
    cout << a << "^3" << " - " << b << "^3 = " << (a-b)*(a*a + a*b + b*b) << "\n";
    cout << a << "^3" << " + " << b << "^3 = " << (a+b)*(a*a - a*b + b*b) << "\n";
    cout << a << "^3" << " + " << b << "^3 = " << (a+b)*(a+b)*(a+b) - 3*a*b*(a+b) << "\n";

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
rehanshahid@Muhammads-MacBook-Air 14:10:26 % g++ taskpl.cpp -o taskpl
rehanshahid@Muhammads-MacBook-Air 14:10:26 % ./taskpl
Enter integer a: 5
Enter integer b: 2
(5 + 2)^2 = 49
(5 - 2)^2 = 9
(5 + 2)^3 = 343
(5 - 2)^3 = 27
5^2 - 2^2 = 21
5^3 - 2^3 = 117
5^3 - 2^3 = 117
5^3 + 2^3 = 133
5^3 + 2^3 = 133
rehanshahid@Muhammads-MacBook-Air 14:10:26 %
```